Measuring and Improving Consistency in Pretrained Language Models

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Abstract

Consistency of a model — that is, the invariance of its behavior under meaning-preserving alternations in its input — is a highly desirable property in natural language processing. In this paper we study the question: Are Pretrained Language Models (PLMs) consistent with respect to factual knowledge? To this end, we create PARAREL🤘, a high-quality resource of cloze-style query English paraphrases. It contains a total of 328 paraphrases for 38 relations. Using PARAREL🤘, we show that the consistency of all PLMs we experiment with is poor – though with high variance between relations. Our analysis of the representational spaces of PLMs suggests that they have a poor structure and are currently not suitable for representing knowledge robustly. Finally, we propose a method for improving model consistency and experimentally demonstrate its effectiveness.[1]

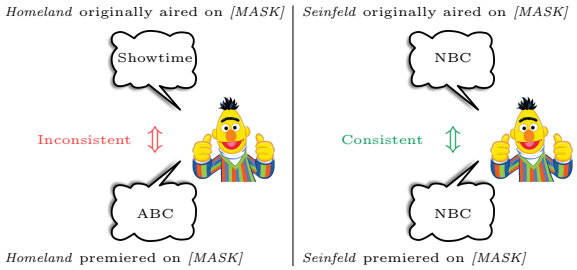

Figure 1: Overview of our approach. We expect that a consistent model would predict the same answer for two paraphrases. In this example, the model is inconsistent on the *Homeland* and consistent on the *Seinfeld* paraphrases.

1 Introduction

Pretrained Language Models (PLMs) are large neural networks that are used in a wide variety of NLP tasks. They operate under a pretrain-finetune paradigm: models are first *pretrained* over a large text corpus and then *finetuned* on a downstream task. PLMs are thought of as good language encoders, supplying basic language understanding capabilities that can be used with ease for many downstream tasks.

A desirable property of a good language understanding model is *consistency*: the ability to make consistent decisions in semantically equivalent contexts, reflecting a systematic ability to generalize in the face of language variability.

Examples of consistency include: predicting the same answer in question answering and reading comprehension tasks regardless of paraphrase (Asai and Hajishirzi, 2020); making consistent assignments in coreference resolution (Denis and Baldridge, 2009; Chang et al., 2011); or making summaries factually consistent with the original document (Kryscinski et al., 2020). While consistency is important in many tasks, nothing in the training process explicitly targets it. One could hope that the unsupervised training signal from large corpora made available to PLMs such as BERT or RoBERTa (Devlin et al., 2019; Liu et al., 2019) is sufficient to induce consistency and transfer it to downstream tasks. In this paper, we show that this is not the case.

The recent rise of PLMs has sparked a discussion about whether these models can be used as Knowledge Bases (KBs) (Petroni et al., 2019, 2020; Davison et al., 2019; Peters et al., 2019; Jiang et al., 2020; Roberts et al., 2020). Consistency is a key property of KBs and is particularly important for automatically constructed KBs. One of the biggest appeals of using a PLM as a KB is that we can query it in natural language – instead of relying on a specific KB schema. The expectation is that PLMs abstract away from language and map queries in natural language into meaningful representations

[1]The code and resource are available at: https://github.com/yanaiela/pararel.

such that queries with identical intent but different language forms yield the same answer. For example, the query "*Homeland* premiered on *[MASK]*" should produce the same answer as "*Homeland* originally aired on *[MASK]*". Studying inconsistencies of PLM-KBs can also teach us about the organization of knowledge in the model or lack thereof. Finally, failure to behave consistently may point to other representational issues such as the similarity between antonyms and synonyms (Nguyen et al., 2016), and overestimating events and actions (reporting bias) (Shwartz and Choi, 2020).

In this work, we study the consistency of factual knowledge in PLMs, specifically in Masked Language Models (MLM) – these are PLMs trained with the MLM objective (Devlin et al., 2019; Liu et al., 2019), as opposed to other strategies such as standard language modeling (Radford et al., 2019) or text-to-text (Raffel et al., 2020). We ask: Is the factual information we extract from PLMs invariant to paraphrasing? We use zero-shot evaluation since we want to inspect models directly, without adding biases through finetuning. This allows us to assess how much consistency was acquired during pretraining and to compare the consistency of different models. Overall, we find that the consistency of the PLMs we consider is poor, although there is a high variance between relations.

We introduce PARAREL🤘, a new benchmark that enables us to measure consistency in PLMs (§3), by using factual knowledge that was found to be partially encoded in them (Petroni et al., 2019; Jiang et al., 2020). PARAREL🤘 is a manually curated resource that provides patterns – short textual prompts – that are paraphrases of one another, with 328 paraphrases describing 38 binary relations such as *X born-in Y*, *X works-for Y* (§4). We then test multiple PLMs for knowledge consistency, i.e., whether a model predicts the same answer for all patterns of a relation. Figure 1 shows an overview of our approach. Using PARAREL🤘, we probe for consistency in four PLM types: BERT, BERT-whole-word-masking, RoBERTa and ALBERT (§5). Our experiments with PARAREL🤘 show that current models have poor consistency, although with high variance between relations (§6).

Finally, we propose a method that improves model consistency by introducing a novel consistency loss (§8). We demonstrate that trained with this loss, BERT achieves better consistency performance on unseen relations. However, more work is required to achieve fully consistent models.

2 Background

There has been significant interest in analyzing how well PLMs (Rogers et al., 2020) perform on linguistic tasks (Goldberg, 2019; Hewitt and Manning, 2019; Tenney et al., 2019; Elazar et al., 2021), commonsense (Forbes et al., 2019; Da and Kasai, 2019; Zhang et al., 2020) and reasoning (Talmor et al., 2020; Kassner et al., 2020), usually assessed by measures of accuracy. However, accuracy is just one measure of PLM performance (Linzen, 2020). It is equally important that PLMs do not make contradictory predictions (cf. Figure 1), a type of error that humans rarely make. There has been relatively little research attention devoted to this question, i.e., to analyze if models behave *consistently*. One example concerns negation: Ettinger (2020) and Kassner and Schütze (2020) show that models tend to generate facts and their negation, a type of inconsistent behavior. Ravichander et al. (2020) propose paired probes for evaluating consistency. Our work is broader in scope, examining the consistency of PLM behavior across a range of factual knowledge types and investigating how models can be made to behave more consistently.

Consistency has also been highlighted as a desirable property in automatically constructed KBs and downstream NLP tasks. We now briefly review work along these lines.

Consistency in knowledge bases (KBs) has been studied in theoretical frameworks in the context of the satisfiability problem and KB construction, and efficient algorithms for detecting inconsistencies in KBs have been proposed (Hansen and Jaumard, 2000; Andersen and Pretolani, 2001). Other work aims to quantify the degree to which KBs are inconsistent and detects inconsistent statements (Thimm, 2009, 2013; Muiño, 2011).

Consistency in question answering was studied by Ribeiro et al. (2019) in two tasks: visual question answering (Antol et al., 2015) and reading comprehension (Rajpurkar et al., 2016). They automatically generate questions to test the consistency of QA models. Their findings suggest that most models are not consistent in their predictions. In addition, they use data augmentation to create more robust models. Alberti et al. (2019) generate new questions conditioned on context and answer from a labeled dataset and by filtering answers that do not provide a consistent result with the origi-

nal answer. They show that pretraining on these synthetic data improves QA results. Asai and Hajishirzi (2020) use data augmentation that complements questions with symmetricity and transitivity, as well as a regularizing loss that penalizes inconsistent predictions. Kassner et al. (2021b) propose a method to improve accuracy and consistency of QA models by augmenting a PLM with an evolving memory that records PLM answers and resolves inconsistency between answers.

Work on **consistency in other domains** includes (Du et al., 2019) where prediction of consistency in procedural text is improved. Ribeiro et al. (2020) use consistency for more robust evaluation. Li et al. (2019) measure and mitigate inconsistency in natural language inference (NLI), and finally, Camburu et al. (2020) propose a method for measuring inconsistencies in NLI explanations (Camburu et al., 2018).

3 Probing PLMs for Consistency

In this section, we formally define consistency and describe our framework for probing consistency of PLMs.

3.1 Consistency

We define a model as *consistent* if, given two *cloze-phrases* such as "*Seinfeld* originally aired on *[MASK]*" and "*Seinfeld* premiered on *[MASK]*" that are *quasi-paraphrases*, it makes non-contradictory predictions[2] on N-1 relations over a large set of entities. A *quasi-paraphrase* – a concept introduced by Bhagat and Hovy (2013) – is a more fuzzy version of a paraphrase. The concept does not rely on the strict, logical definition of paraphrase and allows to operationalize concrete uses of paraphrases. This definition is in the spirit of the RTE definition (Dagan et al., 2005), which similarly supports a more flexible use of the notion of entailment. For instance, a model that predicts *NBC* and *ABC* on the two aforementioned patterns, is not consistent, since these two facts are contradictory. We define a *cloze-pattern* as a cloze-phrase that expresses a relation between a subject and an object. Note that consistency does not require the answers to be factually correct. While correctness is also an important property for KBs, we view it as a separate objective and measure it independently. We use the terms *paraphrase* and *quasi-paraphrase* interchangeably.

Many-to-many (N-M) relations (e.g. *shares-border-with*) can be consistent even with different answers (given they are correct). For instance, two patterns that express the *shares-border-with* relation and predict *Albania* and *Bulgaria* for *Greece* are both correct. We do not consider such relations for measuring consistency. However, another requirement from a KB is *determinism*, i.e., returning the results in the same order (when more than a single result exists). In this work, we focus on consistency, but also measure determinism of the models we inspect.

3.2 The Framework

An illustration of the framework is presented in Figure 2. Let D_i be a set of subject-object KB tuples (e.g., <*Homeland*, *Showtime*>) from some relation r_i (e.g., *originally-aired-on*), accompanied with a set of *quasi-paraphrases* cloze-patterns P_i (e.g., *X* originally aired on *Y*). Our goal is to test whether the model consistently predicts the same object (e.g., *Showtime*) for a particular subject (e.g., *Homeland*).[3] To this end, we substitute *X* with a subject from D_i and *Y* with *[MASK]* in all of the patterns P_i of that relation (e.g., *Homeland* originally aired on *[MASK]* and *Homeland* premiered on *[MASK]*). A consistent model must predict the same entity.

Restricted Candidate Sets Since PLMs were not trained for serving as KBs, they often predict words that are not KB entities; e.g., a PLM may predict, for the pattern "*Showtime* originally aired on *[MASK]*", the noun 'tv' – which is also a likely substitution for the language modeling objective, but not a valid KB fact completion. Therefore, following (Xiong et al., 2020; Ravichander et al., 2020; Kassner et al., 2021a), we restrict the PLMs' output vocabulary to the set of possible gold objects for each relation from the underlining KB. For example, in the *born-in* relation, instead of inspecting the entire vocabulary of a model, we only keep objects from the KB, such as *Paris*, *London*, *Tokyo*, etc.

Note that this setup makes the task easier for the

[2] We refer to *non-contradictory predictions* as predictions that, as the name suggest, do not contradict one another. For instance, predicting as the birth place of a person two difference cities is considered to be contradictory, but predicting a city and its country, is **not**.

[3] Although it is possible to also predict the subject from the object, in the cases of N-1 relations more than a single answer would be possible, making it impossible to test for consistency, but determinism.